

AstroFusion: A GAN-Augmented Approach for Exoplanet Detection

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Abstract—Humanity's exploration of the cosmos spans billions of years, prompting contemplation of our position in the universe. This study advocates for proactive communication with potential intelligent species, driven by uncertainty about their existence. The focus is on exploring Generative Adversarial Networks (GANs) to enhance exoplanet detection through data augmentation. Utilizing the extensive Kepler dataset, the study evaluates the accuracy of conventional machine learning techniques: Logistic Regression, Decision Trees, and Random Forests. (RF)—trained on original data. Subsequently, it assesses the effectiveness of models trained solely on GAN-generated synthetic data and compares their performance with that of models trained on original data. Lastly, it merges original and synthetic data to create a composite dataset, evaluating the ML model's accuracy. The results indicate that models trained solely on synthetic data achieve comparable accuracy to those trained on original data. Furthermore, the combined dataset yields improved accuracy—87%, 95%, and 96% in Logistic Regression, Decision Tree, and RF, respectively—highlighting the potential of GAN-based data augmentation in enhancing exoplanet detection. This study advances space exploration by integrating technology with astronomical research, facilitating more precise and reliable exoplanet detection methods. Leveraging GANs for data augmentation promises novel insights and discoveries about the universe.

Keywords - Kepler, Exoplanet, GAN, Logistic Regression, Decision Tree, Random Forest

1. INTRODUCTION

The quest to explore exoplanets, celestial bodies orbiting stars beyond our solar system, has captivated astronomers and space enthusiasts for decades [1], [2]. With each discovery, our understanding of the universe expands, providing significant understanding into the rich array of planetary systems existing beyond our solar system. The history of exoplanet discovery traces back to the early 1990s, when astronomers Aleksander Wolszczan and Dale Frail made the groundbreaking observation of two planets orbiting a pulsar, marking a significant milestone in astronomical research [2], [3].

The exploration of exoplanets holds immense importance for several reasons. Firstly, it presents an exceptional chance to unravel mysteries surrounding the origins and development of planetary systems. Through examination of exoplanetary systems exhibiting diverse compositions, sizes, and orbital dynamics,

astronomers can refine existing hypotheses on the formation and migration of planets, illuminating the processes shaping planetary systems across the cosmos [4]. Moreover, the quest for exoplanets encompasses the pursuit of habitable worlds—planets with the potential to support life as we understand it. Discovering potentially habitable exoplanets represents a crucial step towards answering the fundamental inquiries surrounding extraterrestrial existence and Earth's singularity within the cosmos. [5].

Furthermore, the study of exoplanets offers valuable insights into planetary diversity and habitability factors. The discovery of exoplanets spanning from gas giants to rocky terrestrial worlds challenges conventional understandings of planetary formation and evolution, enriching our knowledge of the vast cosmic landscape. By examining exoplanetary atmospheres and surface conditions, scientists gain crucial data for understanding planetary climates and assessing the potential for hosting life beyond Earth [6].

Moreover, exoplanet research plays a crucial role in enhancing comprehension of planetary dynamics, including the intricate interplay of tidal forces, planetary migration, and interactions within multi-planet systems. Recent years have seen a revolution in the field of astronomy due to the emergence of machine learning, offering innovative solutions for analyzing vast datasets and uncovering hidden patterns [7]. Among the myriad ML approaches, in recent years, Generative Adversarial Networks (GANs) have become increasingly recognized as a potent tool for creating synthetic data that closely mimics real-world observations [8]. Leveraging GANs for data augmentation presents a hopeful pathway for improving both the precision and efficacy of machine learning models in tasks related to exoplanet detection.

This study aims to explore the potential of GAN-based data augmentation in improving the performance of ML models trained for exoplanet detection. To accomplish this objective, the Kepler cumulative dataset, which comprises extensive records of stellar light curves gathered by the Kepler Space Telescope, acts as the cornerstone dataset [4], [5], [9], [10], [11]. This dataset serves as the cornerstone for training ML models and evaluating

their performance against both original and synthetic datasets generated by GANs.

Through a series of meticulously designed experiments, ML models are trained using both the original Kepler dataset and synthetic datasets. The methodology involves rigorous evaluation and comparison of the accuracies achieved by these models, providing valuable insights into the efficacy of GAN-based data augmentation for enhancing exoplanet detection capabilities.

The research project is structured into multiple sections to offer a thorough grasp of the subject matter. The study commences with an overview of exoplanets, delving into their discovery, characteristics, and significance in astronomy. Subsequently, the diverse techniques employed for exoplanet detection encompass the transit method, radial velocity method, direct imaging, and gravitational microlensing.

Furthermore, the introduction and elucidation of Kepler as well as TESS missions shed light on their invaluable contributions to exoplanet research and the datasets they provide [12], [13], [14], [15]. A detailed description of the Kepler dataset follows, elucidating its sources, characteristics, and relevance to the study. The approach, which combines ML techniques with the transit method, is delineated, along with the methodology for generating synthetic data using GANs.

In subsequent sections, the model training process, alternative approaches, and a comparative analysis of models trained on both original and synthetic data are expounded upon. This culminates in the presentation of results and a comprehensive discussion, drawing conclusions from the findings and exploring avenues for future research and development in this dynamic field. Ultimately, the study emphasizes the transformative potential of GAN-based data augmentation in advancing exoplanet detection methodologies.

2. METHODS OF EXOPLANET DETECTION

Exoplanet detection encompasses various techniques, each with distinct observational principles and applications. The radial velocity method, sometimes referred to as the Doppler or wobble method, is widely utilized in research projects to detect exoplanets [1], [16]. This method depends on measuring minute changes in a star's spectral lines induced due to the gravitational attraction of an orbiting planet. As a planet orbits around its host star, it induces a regular wobble in the star's movement, which can be observed through variations in the star's radial velocity. This method offers significant insights into the mass and orbital characteristics of detected exoplanets.

Another approach, direct imaging, involves capturing images of exoplanets directly using telescopes equipped with advanced imaging technologies [17]. This method is particularly effective for large, young exoplanets emitting their own light and located at significant distances from the stars they orbit. Direct imaging enables detailed investigation of exoplanetary atmospheres and surface features, providing insights into their compositions and properties.

Gravitational microlensing is another technique employed in exoplanet detection, relying on the gravitational bending and amplification of light from a background star due to a foreground object, such as an exoplanet, passing in front of it. By monitoring these transient brightening, astronomers can infer the presence of exoplanets orbiting the foreground object. This approach is especially adept at identifying small exoplanets situated far from their host stars. [18].

Astrometry involves precise measurement of the positions and motions of stars over time to detect the gravitational influence of orbiting exoplanets. By monitoring subtle changes in a star's position relative to background stars, astronomers can infer the presence of exoplanets and estimate their orbital parameters. Astrometry is particularly effective for detecting massive exoplanets orbiting nearby stars.

Investigations into exoplanet detection primarily focus on the transit method, a sensitive technique involving the monitoring of periodic changes in the brightness of a star when an exoplanet passes in front of it. This method offers valuable insights into the existence and attributes of exoplanets, contributing significantly to our understanding of distant planetary systems. The transit depth, represented by equation 1, quantifies the relative decrease in brightness observed during a planetary transit event, encapsulating the essence of the transit method.

$$D = \left(\frac{R_p}{R_s}\right)^2 \quad (1)$$

Where R_p signifies the radius of the exoplanet and R_s represents the radius of the host star.

In addition to ground-based observations, research considers data from specialized astronomical platforms operating beyond the confines of traditional observatories. These platforms, associated with notable missions, provide unique perspectives on celestial phenomena. By incorporating data from such platforms, the study aims to enhance understanding of exoplanetary systems and their complexities.

3. SATELLITES: KEPLER AND TESS

Exploration now shifts to the pivotal role of satellites in advancing the understanding of exoplanetary systems. Specifically, delving into the contributions of two prominent missions: Kepler and TESS. These space-based observatories have revolutionized the field of exoplanet research by providing unparalleled capabilities for detecting and studying distant planets orbiting other stars.

Kepler, sent into orbit by NASA in 2009, the mission was crafted with the purpose of surveying a specific region of the sky in search of exoplanets using the transit method [19], [20]. Over its nearly decade-long mission, Kepler uncovered numerous exoplanets, spanning from compact rocky planets to massive gas giants, and significantly expanded the knowledge of planetary diversity in the Milky Way galaxy [21], [22]. Its precise measurements of exoplanet transits have enabled scientists to determine the size, orbital period, and even atmospheric properties of these distant worlds.

TESS, short for Transiting Exoplanet Survey Satellite, launched in 2018 with the aim of continuing and expanding upon Kepler's

groundbreaking discoveries [23], [24], [25]. Unlike Kepler, which focused on a single patch of sky, TESS conducts a survey of the entire sky, systematically observing hundreds of thousands of stars for signs of planetary transits. This all-sky approach allows TESS to discover exoplanets around a wide range of stars, including nearby bright ones suitable for follow-up observations with other telescopes. Research on TESS has brought significant insights to exoplanet detection.

Together, Kepler and TESS have transformed the understanding of exoplanetary systems, providing a wealth of data that continues to drive scientific inquiry and exploration. By leveraging the capabilities of these space-based observatories, astronomers have made remarkable strides in characterizing exoplanets and uncovering the rich tapestry of worlds beyond the solar system. In the following sections, delving into the datasets provided by Kepler and TESS and discussing their significance for research on exoplanet detection and characterization.

4. THE KEPLER DATASET

Exoplanet data, drawn from esteemed astronomical databases like the NASA Exoplanet Archive and Kepler mission repository, ensures reliability and authenticity. The Exoplanet Archive offers validated data, including orbital parameters and observational details for thousands of exoplanets, meticulously reviewed by expert astronomers. Similarly, Kepler's dataset, comprising photometric time series and light curves, undergoes rigorous validation, contributing to its credibility. Accessing data from these reputable sources underscores a commitment to robust research in exoplanet detection and exploration.

kepid	kepoi_name	kepler_name	koi_disposition	koi_pdposition	koi_score	koi_fpflag_nt
10797460	K00752.01	Kepler-227 b	CONFIRMED	CANDIDATE	1.000	0
10797460	K00752.02	Kepler-227 c	CONFIRMED	CANDIDATE	0.969	0
10811496	K00753.01	NaN	CANDIDATE	CANDIDATE	0.000	0
10848459	K00754.01	NaN	FALSE POSITIVE	FALSE POSITIVE	0.000	0
10854555	K00755.01	Kepler-664 b	CONFIRMED	CANDIDATE	1.000	0

Figure 1 Kepler Dataset sample

Proposed study works based on the Kepler's Third Law, represented by the equation:

$$P^2 = \frac{4\pi^2}{G(M+m) \cdot a^3} \quad (2)$$

where P is the orbital period, 'a' represents the semi-major axis, 'M' stands for the mass of the central star, 'm' denotes the mass of the orbiting planet, and 'G' represents the gravitational constant, plays a crucial role in our understanding of exoplanetary systems.

Transitioning to analysis, Kepler's dataset provides a foundation, with Kepler's Third Law playing a pivotal role. Data preprocessing involves careful curation and feature selection, filtering out redundant elements to focus on essential parameters. Key attributes like orbital period, transit depth, and stellar characteristics are identified, providing insights crucial for ML model training. Incorporation of metadata enhances the dataset, facilitating the distinction between potential exoplanet candidates and confirmed detections. Meticulous curation and feature selection optimize the ML model's performance for accurate exoplanet detection, leveraging the rich dataset from Kepler.

5. PROPOSED APPROACH

The methodology centers on leveraging synthetic data generated by GANs to enhance the efficacy of ML models in exoplanet detection. Initially, the raw data undergoes preprocessing for improved computational understanding. Unlike previous studies [1], [2], [6], [14], [23], [24], [26], [27], [28], [29], [30], the proposed study includes ML models, encompassing logistic regression, decision trees, and RFs, are trained using this processed dataset. Following this, GANs are employed to generate synthetic data closely resembling the authentic dataset, enriching it for model training. After dataset augmentation, ML models are retrained using both the original and synthetic datasets. This retraining occurs in two scenarios: utilizing solely synthetic data (referred to as the "full GAN dataset") and combining an equal mix of real and synthetic data (termed the "50% GAN dataset and 50% original dataset") shown in figure 2.

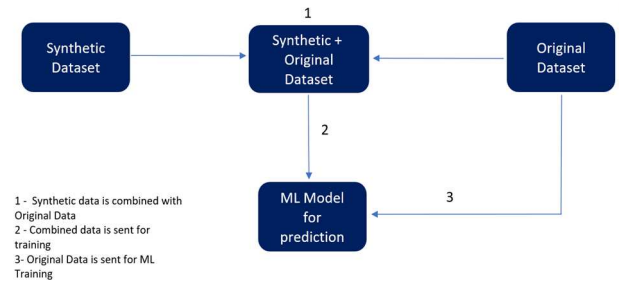


Figure 2 Overall Flow of Dataset Combination

Post-retraining, the model's performance is assessed, considering metrics like accuracy. By comparing the effectiveness of the models across different scenarios, the influence of synthetic data integration on their proficiency in exoplanet detection is assessed.

This systematic analysis aims to elucidate the potential benefits of incorporating GAN-generated synthetic data in improving the capabilities of ML models for exoplanet detection tasks.

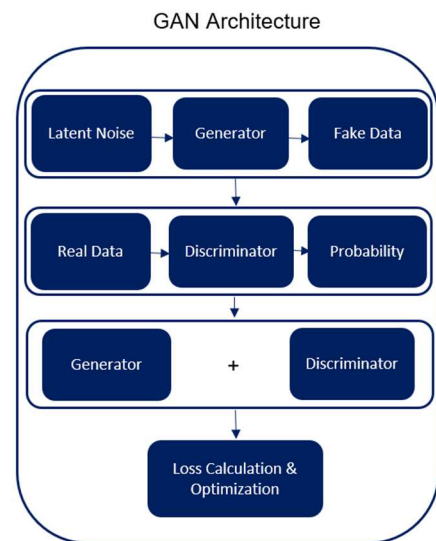


Figure 3 GAN Architecture

Figure 3 illustrates the GAN architecture utilized in this study. The generator produces fake data by incorporating latent noise

into the original dataset. This generated data, along with real data, is then processed in the discriminator block to create an enhanced dataset. Subsequently, the loss calculation is performed on the generated data, followed by optimization.

6. SYNTHETIC DATA GENERATION

In the approach to generating synthetic data, a nuanced strategy tailored to the categorical variables present in the Kepler cumulative dataset was adopted. Instead of training a single GAN on all three categories—Confirmed, Candidate, and False Positive—a more specialized approach was taken. Individual GANs were trained for each categorical variable, as depicted in Figure 4, allowing them to focus exclusively on capturing the unique characteristics of their respective categories.

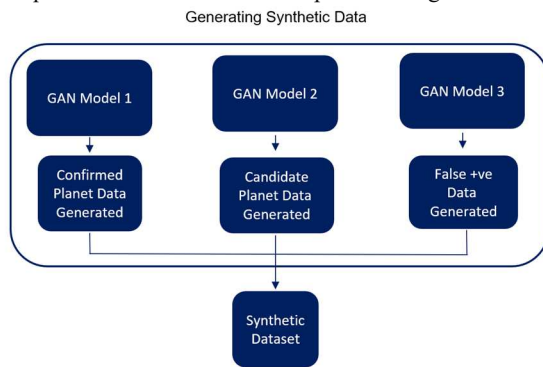


Figure 4 Individual GAN Training

By training separate GANs for each category, the aim was to optimize the synthetic data generation process, ensuring that the generated samples closely resembled the specific features and patterns associated with each category. This targeted approach facilitated fine-tuning of the synthetic data generation process, enhancing the fidelity and relevance of the generated samples for each category. Figures 5 and 6 illustrate the dataset comparison between the original and synthetic dataset of Equilibrium Temperature in two different graphs.

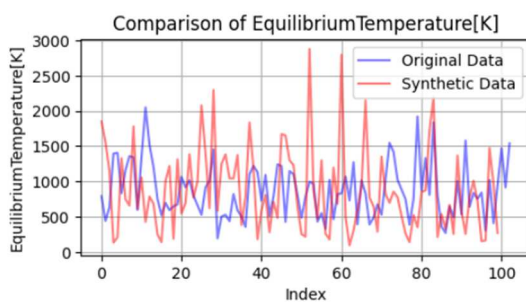


Figure 5 Line Graph Comparison of Equilibrium Temperature (Synthetic and Original)

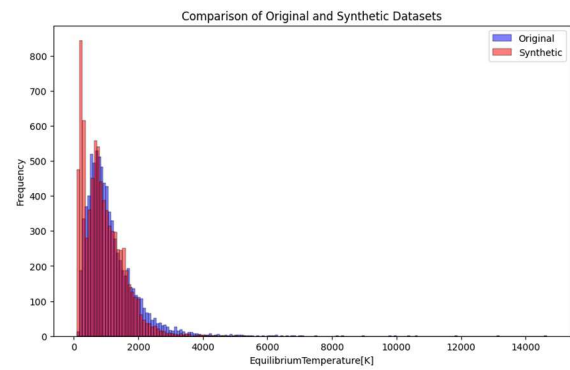


Figure 6 Hist Plot Comparison of Equilibrium Temperature (Synthetic and Original)

Once GANs were trained for each categorical variable, the synthetic data generated by all three GANs were combined to form the final synthetic dataset. This amalgamation of synthetic data from multiple GANs provided a comprehensive dataset that encompassed the diverse characteristics present across all categories.

Overall, the approach of training specialized GANs for individual categories and combining the synthetic data offers a nuanced and effective method for generating synthetic datasets that accurately represent the complexities of the original data across different categorical variables. This tailored approach underscores the commitment to optimizing the synthetic data generation process for improved performance in exoplanet detection tasks.

7. MODEL TRAINING PIPELINE

In the model training process, a structured approach was followed to evaluate the efficacy of utilizing synthetic data in training ML models for exoplanet detection. Initially, multiple ML models—namely logistic regression, decision tree, and RF—were trained using different datasets to assess their performance comprehensively.

Firstly, each ML model was trained exclusively on the original dataset, and its accuracy was evaluated. This initial step provided a baseline for comparison, allowing the performance of the models when trained solely on authentic data to be gauged. Figure 7 explains the training process of both original and synthetic datasets.

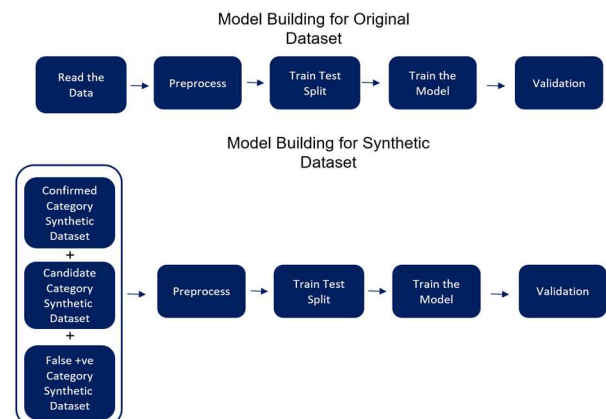


Figure 7 Model Training Process

Subsequently, the training process was repeated, this time utilizing only the synthetic dataset generated by the GAN-based approach. By exclusively training the models on synthetic data, the effectiveness of synthetic data in training ML models for exoplanet detection was assessed.

In addition to training the models separately on original and synthetic datasets, a hybrid approach was explored. The ML models were trained using a balanced combination of original and synthetic data, with each contributing 50% to the training dataset. This hybrid training scheme aimed to leverage the strengths of both authentic and synthetic data, potentially enhancing the models' performance through a diverse training set.

By training multiple models—including logistic regression, decision tree, and RF—using varying datasets, comprehensive insights into the impact of synthetic data on model performance across different ML algorithms were gained. Through systematic evaluation and comparison of model accuracies, the effectiveness of incorporating synthetic data in enhancing the capabilities of ML models for exoplanet detection tasks was elucidated.

8.COMPARATIVE ANALYSIS

In the comparative analysis, the effectiveness of three ML models: Logistic Regression, Decision Tree and RF, across various datasets is assessed. Initially, the accuracy of ML models trained on the original dataset is evaluated, providing a baseline for comparison. Subsequently, the performance of models trained on a fully synthetic dataset generated through GANs is explored, examining any deviations in accuracy and predictive capability compared to the original dataset. Following this, the impact of dataset division is investigated by first splitting the dataset into two parts, without ensuring equal distribution across all categories (i.e., Confirmed, Candidate, False Positive). This preliminary division allows the initial performance of models trained on disparate subsets of the data to be assessed.

Building upon this, the dataset is further refined by meticulously dividing it into two equal parts within each category. This approach ensures that each subset maintains a proportional representation of Confirmed, Candidate, and False Positive samples, thereby providing a more balanced and representative dataset for model training. By training models on these precisely balanced datasets, any nuanced differences in accuracy and performance are discerned, elucidating the effectiveness of synthetic data integration at a category-specific level. Through these comparative analyses, the efficacy of synthetic data augmentation in improving model accuracy and generalization capabilities for exoplanet detection tasks is elucidated. By systematically exploring different dataset compositions and training methodologies, comprehensive insights are provided that can inform future research endeavors and advancements in the field of astronomy and ML.

9. RESULTS AND DISCUSSION

In the final analysis of the study, compelling results derived from a meticulous examination of ML models trained on various

datasets and utilizing diverse methodologies for exoplanet detection are presented. Across all experimental scenarios, significant enhancements in model accuracy and performance upon the integration of synthetic data generated via GANs into the training process are consistently observed.

Specifically, while models trained exclusively on the fully synthetic dataset demonstrate promising performance, their accuracies generally fall short of those achieved by models trained solely on the original dataset. This suggests that synthetic data, while beneficial, may not fully replicate the complexity and variability present in real-world observations.

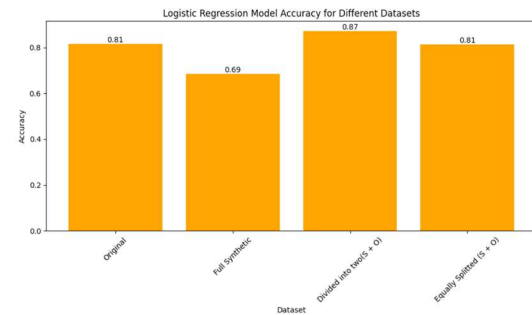


Figure 8 Accuracy Comparison in Logistic Regression

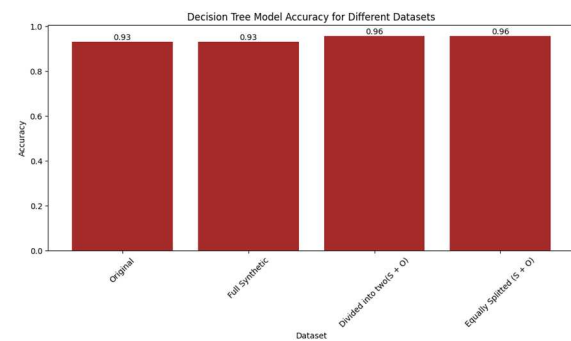


Figure 9 Accuracy Comparison in Decision Tree

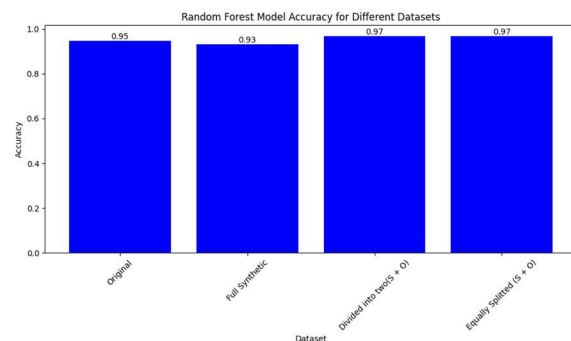


Figure 10 Accuracy Comparison in RF

Furthermore, upon evaluating datasets featuring a combination of original and synthetic data, a consistent trend towards improved model performance as the proportion of synthetic data increases is noted. Models trained on datasets comprising a balanced blend of original and synthetic data consistently exhibit higher accuracies and superior generalization abilities compared to those trained solely on the original dataset.

The accuracy of a classification model is calculated using the equation:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \times 100\% \quad (3)$$

Similarly, the F1-Score, which considers both precision and recall, is given by:

$$F1 = 2X \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} \quad (4)$$

Figure 8 through 15 shows the accuracy and F1-Score comparison of different models namely Logistic Regression, Decision Tree, and RF. Figure 8 illustrates the varying accuracies of logistic regression models trained on different datasets. While the original dataset achieves 81% accuracy, the full synthetic dataset sees a decrease to 69%, with a notable increase to 87% and 81% for randomly and equally split datasets, respectively.

In Figure 9, the accuracy of decision tree models is displayed across different datasets. Notably, both the original and fully synthetic datasets yield a high accuracy of 93%, with further improvements to 96% for both randomly and equally split datasets.

Moving to Figure 10, the accuracy comparison for RF models is depicted. Here, the original dataset achieves a commendable accuracy of 95%, while the full synthetic dataset slightly decreases to 93%. However, both randomly and equally split datasets exhibit impressive accuracies of 97%.

Figure 11 showcases the accuracy comparison of equally split datasets across logistic regression, decision tree, and RF models. Notably, all models achieve high accuracies, with decision tree and RF outperforming logistic regression.

In Figure 12, the F1-score comparison of equally split datasets across the three models is depicted. The decision tree and RF models demonstrate superior F1-scores compared to logistic regression, indicating their better performance in classification tasks.

Figure 13 presents the F1-score comparison in logistic regression, highlighting slight variations across different datasets. While the original and randomly split datasets show relatively higher F1-scores, the full synthetic dataset exhibits a decrease, with the equally split dataset maintaining a consistent F1-score.

Similarly, Figure 14 displays the F1-score comparison in decision tree models, with noteworthy consistency across different datasets. Notably, both the original and randomly split datasets yield higher F1-scores compared to the fully synthetic dataset.

Finally, in Figure 15, the F1-score comparison in RF models is depicted. Consistent with previous observations, the original and randomly split datasets demonstrate higher F1-scores compared to the fully synthetic dataset, while the equally split dataset maintains a high F1-score.

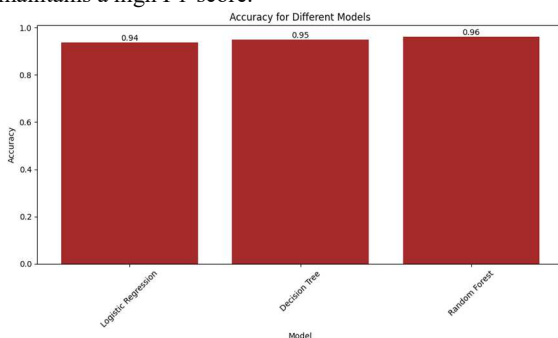


Figure 11 Accuracy of Splitted dataset across three models

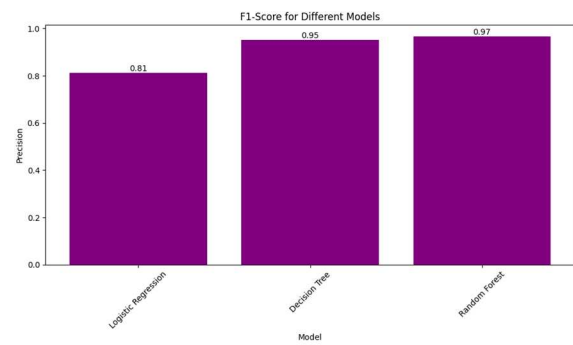


Figure 12 F1- Score of Splitted dataset across three models

In the evaluation of ML models for exoplanet detection, the primary metrics of interest are accuracy and F1 score, providing valuable insights into the overall performance and performance of the models. Accuracy represents the ratio of correctly classified instances to the total instances in the dataset, providing a straightforward measure of model's correctness. On the other hand, the F1 score considers both the precision and recall of the model, offering a balanced assessment of its performance by considering the trade-off between precision and recall. These metrics are fundamental for evaluating the model's capacity to accurately identify exoplanets while minimizing false positives and false negatives.

For a deeper insight into the model's performance, it is essential to consider additional evaluation parameters beyond accuracy and F1 score. One such metric is precision, quantifying the ratio of true positive predictions to all positive predictions made by the model. Precision is especially valuable in situations where minimizing false positives is crucial, such as in medical diagnosis or anomaly detection. Additionally, recall, sometimes referred to as sensitivity, calculates the ratio of true positive predictions identified by the model among all actual positive instances in the dataset. High recall indicates the model's capability to capture all relevant instances of the positive class, making it crucial in situations where detecting all positive cases is paramount.

Moreover, specificity, another important metric, evaluates the model's capacity to accurately identify true negatives among all actual negative instances in the dataset. By considering specificity alongside sensitivity, one can assess the effectiveness of the model across both positive as well as negative classes, providing a more nuanced perspective on its overall effectiveness. Finally, the ROC-AUC metric, which evaluates the area under the Receiver Operating Characteristic curve, provides a comprehensive assessment of the model's capacity to differentiate between the positive and negative instances across different thresholds for classification. Incorporating these additional evaluation parameters enhances comprehending of the model's strengths and weaknesses, facilitating more informed decision-making in model selection and optimization.

Additionally, the analysis underscored the critical role of dataset composition and the distribution of samples across different categories (Confirmed, Candidate, False Positive) in influencing model performance. Models trained on datasets with a well-balanced distribution of samples across all categories consistently outperformed those trained on unbalanced datasets, emphasizing the significance of meticulous dataset preprocessing and management practices.

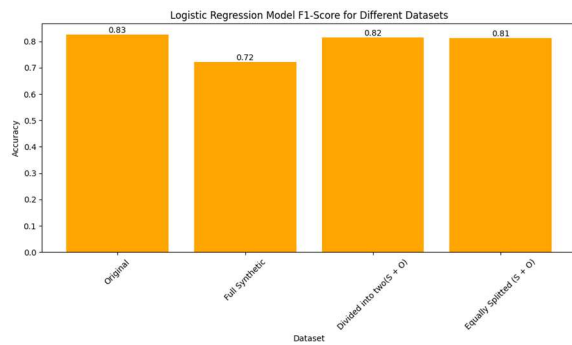


Figure 13 F1-Score derived from Logistic Regression

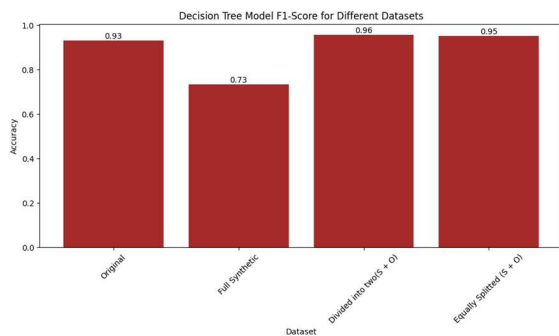


Figure 14 F1-Score derived from Decision Tree

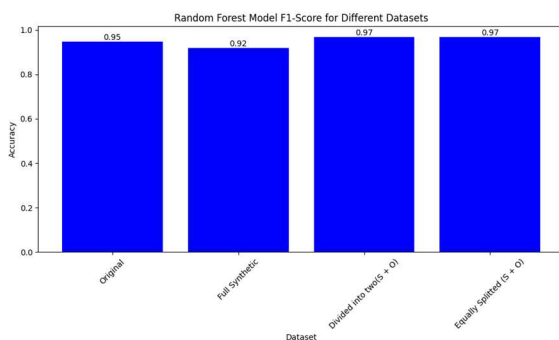


Figure 15 F1-Score of RF

In analyzing the model performances, it is observed that RF initially displayed the highest accuracy (Figure 16).

In the domain of exoplanet detection, the RF algorithm stands out for its remarkable ability to navigate the complexities inherent in exoplanet datasets. RF excels in handling high-dimensional and noisy data, which are common traits of datasets derived from astronomical observations. By employing a large ensemble of decision trees, RF can effectively capture intricate patterns and nonlinear relationships present in the data, facilitating accurate classification of exoplanet candidates.

Moreover, the ensemble nature of RF lends itself well to exoplanet detection tasks by mitigating the risk of overfitting and improving generalization performance. Each decision tree within the ensemble is trained on a random subset of the training data, and predictions are aggregated across multiple trees. This ensemble approach aids in alleviating the impacts of noise and variability in the data, resulting in more robust and reliable predictions. Additionally, RF's capability to handle missing data and outliers further enhances its suitability for analyzing

exoplanet datasets, where data quality may vary due to factors such as instrument noise and observational errors.

Another notable strength of RF is its interpretability, which is crucial for understanding the underlying features and factors contributing to the classification of exoplanet candidates. RF provides insights into feature importance, enabling researchers to identify the most discriminative features for distinguishing between different classes of exoplanets. This interpretability aspect is particularly valuable in the context of exoplanet research, where understanding the physical and astronomical properties of detected exoplanets is essential.

In summary, the RF algorithm offers a potent and versatile approach for analyzing exoplanet datasets, leveraging its robustness, ensemble nature, and interpretability to achieve high accuracy and reliability in exoplanet detection tasks. Its capacity to handle high-dimensional, noisy data makes RF well-suited for extracting meaningful insights from exoplanet datasets and advancing our understanding of the vast diversity of exoplanetary systems in the universe.

However, when synthetic datasets are introduced, Decision Tree exhibits a notable 3% increase in accuracy, compared to the 2% increase observed in RF. This nuanced comparison suggests that while RF may initially have higher accuracy, Decision Tree demonstrates more significant improvement when synthetic data is utilized. Consequently, Decision Tree emerges as a potentially superior model in this context. It's crucial to note that this assessment is solely based on accuracy metrics; other factors such as computational efficiency and model interpretability warrant consideration in determining the optimal model. This analysis underscores the importance of a comprehensive evaluation approach in discerning model performance nuances.

Model	Original Dataset	Synthetic Dataset	Mixed Dataset	Splitted Dataset
Logistic Regression	81%	68%	87%	81%
Decision Tree	92%	88%	95%	95%
Random Forest	94%	91%	96%	96%

Figure 16 Result with accuracy values of different models trained on different datasets.

In general, synthetic data augmentation techniques show promise in enhancing model performance for exoplanet detection. However, further research is needed to optimize their effectiveness and address potential limitations. By continuing to refine these methodologies, the field of exoplanet detection can advance, deepening the understanding of distant planetary systems.

10. CONCLUSION

The exploration of ML algorithms for exoplanet detection, particularly focusing on RF and Decision Tree models, underscores the nuanced dynamics of model performance. RF initially showcased remarkable accuracy, leveraging its ensemble-based approach to handle the intricacies of high-dimensional and noisy exoplanet datasets effectively. However, the introduction of synthetic datasets revealed an intriguing aspect

of model performance. While RF displayed a modest increase in accuracy, Decision Tree exhibited a more substantial improvement, suggesting its potential superiority in certain contexts. This nuanced comparison emphasizes the importance of considering various factors beyond just accuracy, including computational efficiency and model interpretability, when determining the optimal approach for exoplanet detection. Moreover, the promising role of synthetic data augmentation techniques in enhancing model performance underscores the necessity for ongoing research to refine these methodologies and deepen our comprehension of distant planetary systems. Through ongoing refinement and innovation, we can harness the full potential of these techniques to unravel the mysteries of exoplanetary systems.

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